Project Report

How To Set Up A Business Unit For WIREWOUND RESISTORS MANUFACTURING

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Wirewound resistor is a type of passive component in which metal wires are used to reduce or restrict the flow of electric current to a certain level. It is of two types: power wirewound resistor and precision wirewound resistor.

Power wirewound resistor. This is a non-inductive wirewound resistor that operates at high temperature. It is commonly used for high power applications.

Precision wirewound resistor. This operates at low temperature with high accuracy. It is used as a precision resistor in instrumentation because of its high accuracy.

Increasing demand for electronic devices and equipment is driving the increasing growth of the wirewound resistors market

Wirewound resistors are used only for low frequencies, and are not suitable for high frequencies. At high frequencies, these act as inductors. Hence, for high frequencies, non-inductive wirewound resistors are used.

Wirewound resistors find use in almost all major electronic circuits, and are widely used in applications such as telecommunications, computers, audio and video equipment, medical electronic equipment, defence and space, telephone switching systems, transducers instrumentation, current and voltage balancing, and current sensing.

Market potential

There are many units in India that manufacture different types of wirewound resistors, including silicon-coated, ceramic-encased and aluminium-wound resistors. But, there is an incremental growth of electronic industries in the country, which has created further scope for new industries in the field of wirewound resistors.

Increasing demand for electronic devices and equipment is the key factor driving the increasing growth of the wirewound resistors market. Most consumer electronics have inbuilt wirewound resistors to avoid or minimise the flow of excessive electric current. Wirewound resistors provide high protection to electronics equipment by reducing fluctuations in the flow of extra current near their threshold voltage. Home appliances and electronic devices such as cellphones,

Business compliances

In the beginning, as a manufacturer, apply for different registrations and licences from the relevant government authorities. However, it is advisable to check the state law.

1. Register the business with Registrar of Companies (ROC).
2. Apply for trade licence from the local municipal authority.
3. Apply for MSME Udyog Aadhaar for small- and medium-unit operations. This can be registered online.
4. Obtain NOC from Pollution Control Board. First, apply for Consent to Establish and then Consent to Operate.
5. Apply for GST registration.
6. Protect brand name with trademark registration.

Quality control and standards. Since performance of electric equipment depends on the quality of components used, wirewound resistors need to adhere to IS 8909 (Part - 1 to 5) 1978 standard.

RoHS compliance. RoHS stands for Restriction of Hazardous Substances. It impacts the entire electronics industry and many electrical products as well. The original RoHS, also known as Directive 2002/95/EC, originated in the European Union (EU) in 2002 and restricts the use of six hazardous materials found in electrical and electronic products. All applicable products in the EU market July 1, 2006 onwards must pass RoHS compliance.

Directive 2011/65/EU was published in 2011 by the EU, which is known as RoHS-Recast or RoHS 2. RoHS 2 includes a CE-marking directive, with RoHS compliance now being required for CE marking of products. RoHS 2 adds Categories 8 and 9, and has additional compliance record-keeping requirements.

Directive 2015/863 was published in 2015 by the EU, which is known as RoHS 3. RoHS 3 adds four additional restricted substances (phthalates) to the existing list of six.

Any business that sells applicable electrical or electronic products, equipment, sub-assemblies, cables, components or spare parts directly to the countries that mandate RoHS rules, or sells to resellers, distributors or integrators that, in turn, sell products to these countries, is impacted if these utilise any of the restricted 10 substances.

RoHS specifies maximum levels for the following 10 restricted substances. The first six applied to the original RoHS, while the last four were added under RoHS 3.

1. Lead (Pb): <1000ppm
2. Mercury (Hg): <100ppm
3. Cadmium (Cd): <100ppm
4. Hexavalent chromium: (Cr VI) <1000ppm
5. Polybrominated biphenyls (PBB): <1000ppm
6. Polybrominated diphenyl ethers (PBDE): <1000ppm
7. Bis(2-Ethylhexyl) phthalate (DEHP): <1000ppm
8. Benzyl butyl phthalate (BBP): <1000ppm
9. Dibutyl phthalate (DBP): <1000ppm
10. Diisobutyl phthalate (DIBP): <1000ppm